Computational Suitability Statement (CSS)

Project and PI Information:

Project Title: Learning Generalizable Dexterous Manipulation from Human Videos
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Abstract:
Dexterous robot learning remains fundamentally limited by the scarcity of large-scale robot demonstration data. Egocentric human video offers a promising alternative, capturing rich manipulation behavior across diverse objects and interactions. Recent work shows that this signal can be leveraged either through large-scale pretraining or by reducing the embodiment gap via visual editing and inpainting of human hands into robot-compatible representations [1,2]. We aim to extend these advances to dexterous manipulation, where multi-fingered hands introduce a larger embodiment gap and higher-dimensional control. Our project will develop methods that combine edited human video and robot data to pretrain dexterous world and action models, and evaluate whether this improves generalization and planning in dexterous tasks.

Scientific Overview:
Dexterous manipulation remains a core challenge in robot learning due to high-dimensional control, complex contact dynamics, and strong embodiment mismatch between humans and robots [3]. Recent work shows that egocentric human video can provide scalable supervision for dexterous policies, enabling transfer of rich interaction structure beyond what robot data alone can support [4,5]. In parallel, video world models—generative models that predict future observations conditioned on actions—offer a powerful framework for planning, policy evaluation, and data generation in robotics [1,6]. However, it remains unclear whether these approaches can be combined to enable scalable learning of dexterous skills. Existing methods either reduce the embodiment gap through visual editing or retargeting of human demonstrations [2,3,5], or rely on large-scale co-training on human and robot data to learn transferable priors [1,4]. Yet their effectiveness in high-dimensional dexterous settings is still largely unexplored. This project aims to train large-scale video world and action models for dexterous manipulation by leveraging egocentric human video. We will develop methods to extract dexterous interaction structure, bridge embodiment mismatch, and integrate human and robot data. We evaluate whether such pretraining improves policy learning, generalization to novel objects, and test-time planning in dexterous robotic tasks.

Prior Experience:

Our team has extensive experience working with SLURM-managed compute clusters across Stanford's SC infrastructure, Google Cloud Platform, and other heterogeneous environments. We have hands-on experience training large-scale models on diverse, high-volume datasets (e.g., DROID, Open X), and are familiar with distributed multi-GPU workflows required for such workloads. In addition, we have experience provisioning and managing GPU resources in the cloud, including H100 instances for large model fine-tuning.

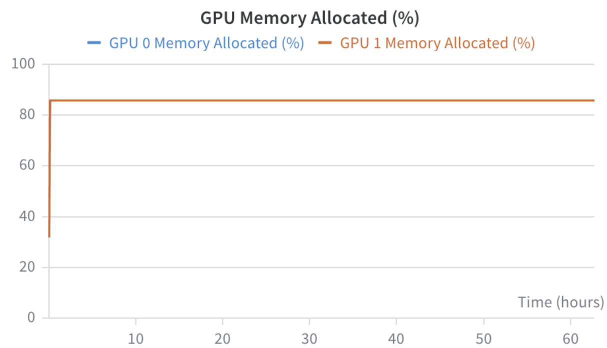
Through this experience, we have established robust practices for running multi-GPU workloads, including careful monitoring of GPU utilization, memory usage, and training throughput to ensure efficient hardware use. We routinely profile jobs at early stages and iteratively tune batch sizes, gradient accumulation, and parallelism strategies before scaling to full experiments.

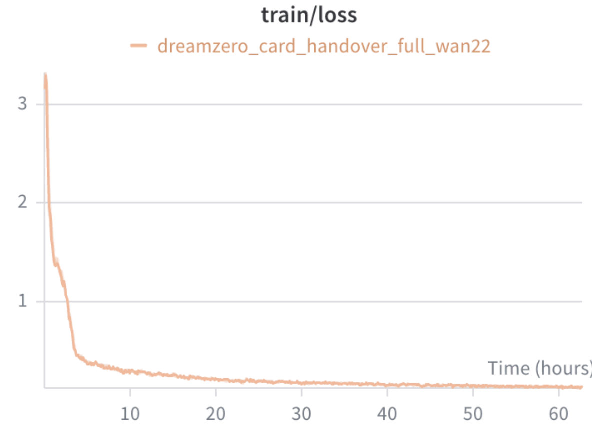
Scaling Studies:

A key aspect of our computational requirements is scaling large model parameters—working across the 3B to 8B range—where our training pipelines require large video datasets processed concurrently in GPU memory, either co-located across multiple GPUs on a single node or distributed across independent nodes. Throughout world model fine-tuning, these distributed processes must routinely communicate intermediate gradients and activations to make forward progress. This high inter-GPU communication regime, combined with the need for multiple concurrent H100 GPUs, underscores our need for a reliable, high-performance cluster with fast interconnects, which Marlowe specifically provides.

Our typical jobs require 2-8 NVIDIA H100 GPUs, with utilization maintained at 80–100% and minimal CPU demands (e.g. about 3% on an a3-highgpu-2g VM on GCP). RAM usage typically ranges from 64-128 GB. Training jobs are fully checkpoint-enabled, with each checkpoint occupying approximately 20 GB, allowing seamless pause and restart.

We demonstrate here efficient GPU utilization in our implementations. Profiling of our world model training runs shows consistent high throughput and effective saturation of the available ~80 GB VRAM on H100 GPUs. These metrics confirm our implementations fully leverage the hardware. Below, we show an logs of an example DreamZero training run on a3-highgpu-2g VM on GCP. The loss curve has reached near convergence after around 3 days of training, and the GPU utilization is high (~86%; we cannot increase batch size further). In the future, for large-scale jobs, we plan to use 8 A100/H100 GPUs for training.





Computational Profile:
Most of our experiments involve training 3–8 B-parameter world action models on 2–8 NVIDIA H100 GPUs. A typical job runs for 12–36 hours on 8 GPUs in a compute-bound setting, while larger pretraining runs scale to multi-node (up to 16 GPUs) with NVLink-enabled communication. We use the already implemented distributed training pipelines for DreamZero (https://dreamzero0.github.io/) to ensure efficient scaling and throughput. During peak phases, we run up to 16 jobs concurrently.

Justification for Marlowe:
Our rationale for requesting Marlowe stems from the inadequacy of available A100/H100 GPUs resources through Google Cloud Platform and our lab servers, which suffer from high contention. Moreover, our lab servers do not provide H100 GPUs. The dedicated resources and priority scheduling on Marlowe would allow faster experiment turnaround and the stable multi-day runtimes our pretraining jobs require, significantly accelerating our research cycles. Moreover, our workflows will benefit significantly from Marlowe's NVLink-enabled GPUs, which reduce communication overhead for gradient synchronization across devices. Provisioning equivalent capacity through cloud providers is cost-prohibitive at our scale.

Technical Requirements & Feasibility:

All software dependencies are open-source and pip-installable. Training relies on the DreamZero framework (publicly available), with standard dependencies including PyTorch, CUDA 12, NCCL, and Hugging Face libraries. No proprietary software, special hardware peripherals, or non-standard kernel configurations are required.

Storage & Data:

Our storage needs primarily consist of scratch space for large-scale video data, intermediate processing, and model checkpoints. Data will be sourced from publicly available egocentric video datasets and dexterous manipulation datasets (e.g., Epic Kitchens, EgoDex, EgoVerse), downloaded directly to cluster storage or transferred via SCP/rsync. Intermediate artifacts include tokenized video shards, hand-pose estimates, inpainted video clips, and latent action representations. We checkpoint models regularly during training (~20 GB per checkpoint), retaining only the most recent and best-performing checkpoints while cleaning up unused data after analysis. Due to the scale of video pretraining and concurrent experiments, we estimate an overall scratch storage requirement of approximately 100 TB.

Impacts and Acknowledgements

This project addresses a key barrier in robot learning: leveraging large-scale human manipulation video despite the embodiment mismatch. Success would enable scalable learning of dexterous manipulation through open-source world models, edited video datasets, and training pipelines. We plan to publish at CoRL 2026 and open-source all code, models, and datasets.

References:

[1] DreamZero: World Action Models are Zero-shot Policies. NVIDIA Research, *arXiv*, 2026.

[2] Masquerade: Learning Dexterous Manipulation from Roboticized Human Videos. ICRA, 2025.

[3] DexMachina: Functional Retargeting for Bimanual Dexterous Manipulation. CoRL, 2025.

[4] EgoScale: Scaling Egocentric Human Video for Dexterous Manipulation. *arXiv*, 2025.

[5] UniDex: A Universal Dexterous Hand Foundation Suite from Egocentric Human Videos. CVPR, 2026.

[6] LingBot-World: An Open-Source Interactive World Simulator. *arXiv*, 2025.